The Æther-flow Interpretation of Relativity

Higher-Derivative Quartic Branch Derivational Bridge-Closure Test

Abstract

The fixed-completion higher-derivative quartic branch has now passed the coefficient-level flat-background exact-closure screen, the bridge-metric matter-coupling gate, the recovery-compatibility gate, the admissible-background theorem route at its current disciplined boundary, the explicit-completion local and coercive small-data stages, the weighted/homogeneous gauge-first repair line, and the same-class asymptotic/nonlinear-stability closure test. The next honest burden is therefore no longer another stability repair. It is derivational: does the surviving branch actually close back onto the exact Einsteinian observer-accessible line, or does it remain only a stable recovery-compatible candidate completion?

This note records that next proof-or-failure test in the narrowest disciplined form. It defines the *bridge correction tensor* $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ by the reduced observer equation

$$G_{\mu\nu}[g] = 8\pi G_N\, T^{\mathrm{matter}}_{\mu\nu} + \mathcal{C}^{\mathrm{br}}_{\mu\nu},$$

where $g_{\mu\nu} = \Xi^* \mathfrak{g}_{AB}$ is the single operational metric already fixed in the active exact-closure line. The current manuscript sequence already supplies four strong restrictions on $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$: it is metric-mediated rather than a direct matter coupling to extra substrate variables; at the present flat-branch quadratic scope it contributes no unsuppressed k^0 or k^2 observer-accessible deformation; on the surviving fixed completion its small-data dynamics are globally controlled in the same weighted/homogeneous class; and along that route no first genuine asymptotic obstruction appears.

Those results are favorable but not yet a derivation. Smallness, infrared suppression, and same-class nonlinear stability do not by themselves imply exact closure. The remaining derivational question is therefore exact and sharp: does the bridge correction tensor vanish identically, or reduce to a pure gauge/constraint-exact term with no observer-accessible content, on the surviving branch in the intended recovery regime? This note shows that this condition is now the precise remaining bridge-closure criterion. If it is proved, the surviving quartic branch enters the active derivational line. If it fails, the branch remains a stable recovery-compatible candidate but not a completed substrate derivation of Einsteinian gravity.

1. Introduction

The active manuscript sequence of Æther-flow Interpretation of Relativity now has both a fixed observer-accessible exact-relativistic line and a much sharper candidate derivational branch than before. The observer-accessible line is already fixed by adoption: Einsteinian gravity with universal matter coupling through a single operational metric. The derivational side is no longer only a schematic bridge either. The higher-derivative quartic branch has survived the flat-background exact-closure screen, preserved single-metric matter coupling at the present action-level and quadratic

scope, remained compatible with the active recovery line in its stated infrared regime, and now passed the same-class asymptotic/nonlinear-stability route for one explicit fixed completion.

That changes the next burden again. The remaining question is no longer whether the branch is immediately destabilized by the old lapse obstruction, by the first unit-lapse trace/auxiliary failure, or by the weighted/homogeneous post-local dispersive argument. Those steps are now on record. The next question is derivational: does the surviving branch close back onto the exact Einsteinian observer equation, or does some residual observer-accessible bridge correction survive even on the now-stable branch?

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays within that boundary. It does not claim that Einsteinian gravity has already been derived from substrate microphysics. It does not claim that the surviving quartic branch already proves the exact observer equation in full nonlinear generality. It performs the next narrower test: define the exact remaining bridge-closure burden now that the surviving fixed-completion route is both recovery-compatible and same-class asymptotically stable.

2. What the Current Sequence Already Controls

The current branch data needed for the present note are already fixed elsewhere in the sequence.

Definition 1 (Current surviving quartic branch data)**.** *The* current surviving quartic branch data *consist of:*

1. *the bridge metric*
$$\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B \tag{1}$$
and the observer metric
$$g_{\mu\nu} = \Xi^* \mathfrak{g}_{AB} \tag{2}$$

2. *the universal matter route*
$$S_{\text{matter}}[\Xi^* \mathfrak{g}, \psi] = S_{\text{matter}}[g, \psi] \tag{3}$$

3. *the higher-derivative collapsed-scalar quartic branch with*
$$\kappa_S = \mu_{SI} = \mu_S^2 = 0, \qquad m_S^2 > 0, \qquad \lambda_4 > 0, \tag{4}$$
together with positive-definite Q-sector and ordered-flow auxiliary blocks

4. *the explicit fixed completion $S_{\min,4}$ and its unit-lapse weighted/homogeneous same-class asymptotic closure route.*

Proposition 1 (Current sequence restrictions on the surviving branch)**.** *At the present disciplined scope, the active manuscript sequence already yields all of the following for the branch data of Definition 1:*

1. ***single-metric matter coupling:*** *matter and light couple only through $g_{\mu\nu}$*
2. ***flat-branch k^0/k^2 exact-closure compatibility:*** *the reduced observer-accessible scalar correction contributes no unsuppressed k^0 or k^2 term*
3. ***recovery compatibility:*** *the branch preserves the benchmark null cone, proper-time rule, and local-SR structure in the stated long-wavelength non-ultrasoft infrared regime*
4. ***same-class nonlinear control:*** *on the fixed-completion unit-lapse weighted/homogeneous route, the small-data dynamics close globally with a mild transport-dressed asymptotic profile and no first genuine same-class obstruction.*

Proof. Item (1) is the matter-coupling gate result. Item (2) is the flat-background higher-derivative consistency result together with the recovery-compatibility note, which shows that the only recorded observer-accessible quadratic branch correction is quartic in spatial momentum. Item (3) is exactly the content of the recovery-compatibility note. Item (4) is exactly the content of the weighted/homogeneous local, coercive-bootstrap, decay/integrability, and asymptotic/nonlinear-stability closure notes on the fixed completion $S_{\min,4}$. □

Remark 1. *Proposition 1 is strong enough to rule out several earlier routes to failure, but it is intentionally weaker than a derivation. It does not say that the branch correction vanishes identically. It says only that every currently recorded observer-accessible correction is tightly constrained and dynamically controlled on the surviving branch.*

3. Bridge Correction Tensor and Exact Closure on the Surviving Branch

The exact derivational burden is best stated at the level of the reduced observer equation.

Definition 2 (Bridge correction tensor)**.** *For the surviving quartic branch, define the* bridge correction tensor $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ *by the reduced observer equation*

$$G_{\mu\nu}[g] = 8\pi G_N\, T^{\mathrm{matter}}_{\mu\nu} + \mathcal{C}^{\mathrm{br}}_{\mu\nu}[g;\mathcal{A},U,S,Q]. \tag{5}$$

Here $G_{\mu\nu}[g]$ is the Einstein tensor of the single operational metric $g_{\mu\nu}$, and $T^{\mathrm{matter}}_{\mu\nu}$ is the stress tensor derived from the matter route (3)*. The tensor $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ is defined after eliminating the nonmetric substrate variables through their branch equations and constraint structure.*

Remark 2. *Definition 2 is not a claim that $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ is already known explicitly at full nonlinear order. It only records the exact object whose vanishing or nonvanishing decides derivational closure on the surviving branch.*

Definition 3 (Pure gauge/constraint-exact remainder)**.** *The bridge correction tensor is said to be* pure gauge/constraint exact *on a branch if there exist a one-form Y_μ and a collection of reduced constraint quantities $\mathcal{H}$, $\mathcal{M}_\mu$, $\mathcal{A}_a$ such that*

$$\mathcal{C}^{\mathrm{br}}_{\mu\nu} = \nabla_{(\mu} Y_{\nu)} + \mathcal{P}_{\mu\nu}\mathcal{H} + \mathcal{P}_{\mu\nu}{}^{\rho}\mathcal{M}_\rho + \sum_a \mathcal{P}^{(a)}_{\mu\nu}\mathcal{A}_a, \tag{6}$$

where every term on the right vanishes on solutions of the reduced observer-accessible system modulo the usual diffeomorphism freedom of $g_{\mu\nu}$.

Definition 4 (Derivational bridge closure on the surviving branch)**.** *The surviving quartic branch is said to satisfy* derivational bridge closure *in a stated regime if, after branch reduction, either*

$$\mathcal{C}^{\mathrm{br}}_{\mu\nu} \equiv 0 \tag{7}$$

or $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ is pure gauge/constraint exact in the sense of Definition 3.

Remark 3. *Definition 4 is the correct exact-closure condition at the observer-equation level. Recovery compatibility and nonlinear stability are necessary preparatory results, but neither is by itself sufficient. Exact derivation requires the residual branch correction to disappear from the observer-accessible equation itself, not merely to remain small or infrared-suppressed.*

4. Reduction of the Remaining Derivational Burden

The point of the present note is that the current repository is now strong enough to isolate the remaining burden to the tensor $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ itself.

Proposition 2 (What has already been removed from the derivational burden)**.** *For the surviving quartic branch, the current sequence already removes the following as immediate reasons to reject derivational closure:*

1. *a direct low-energy matter coupling to extra substrate variables*
2. *an unsuppressed k^0 or k^2 weak-field deformation at the present flat-branch quadratic scope*
3. *a first same-class asymptotic instability on the fixed completion $S_{\mathrm{min},4}$*
4. *a first immediate need for a different gauge, deeper completion, or auxiliary redesign before testing the observer equation itself.*

Proof. Item (1) is excluded by the bridge-metric matter-coupling gate. Item (2) is excluded by the higher-derivative consistency and recovery-compatibility notes. Item (3) is excluded by the weighted/homogeneous asymptotic/nonlinear-stability closure note. Item (4) follows because that same-class route closes positively rather than failing at the asymptotic stage. □

Theorem 1 (Exact proof-or-failure criterion for the next derivational step)**.** *Assume the branch data of Definition 1 and the currently recorded results summarized in Proposition 1. Then the remaining derivational question for the surviving quartic branch is equivalent to the following exact proof-or-failure test:*

> *Determine whether the reduced observer equation* (5) *satisfies the bridge-closure condition of Definition 4 in the intended recovery regime.*

More precisely:

1. *if $\mathcal{C}^{\mathrm{br}}_{\mu\nu} \equiv 0$ or is pure gauge/constraint exact, then the surviving quartic branch closes onto the exact Einsteinian observer-accessible line in that regime*

2. *if $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ contains any nonzero observer-accessible remainder not of the form* (6)*, then the branch remains only a stable recovery-compatible candidate completion rather than a completed substrate derivation.*

Proof. By Proposition 2, the current sequence has already ruled out the earlier direct matter-coupling failure, the earlier low-order weak-field deformation, and the earlier first stability-level obstruction on the fixed completion. Therefore the remaining unresolved issue is no longer whether the branch can be kept alive mathematically at the current disciplined scope. The unresolved issue is whether any residual observer-accessible branch correction survives in (5).

If $\mathcal{C}^{\mathrm{br}}_{\mu\nu} \equiv 0$, then (5) is exactly the Einstein equation with universal matter coupling, so the branch closes onto the exact relativistic line. If $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ is pure gauge/constraint exact, then it carries no observer-accessible content on reduced solutions and the same conclusion follows. Conversely, if a nonzero observer-accessible remainder survives, then the branch does not derive the exact adopted observer equation even though it may remain recovery-compatible and dynamically stable. This is exactly the distinction between derivation and stable candidate dynamics in the active manuscript sequence. □

Corollary 1 (Next manuscript burden)**.** *The next honest manuscript after the present note is not another same-class stability repair. It is the explicit evaluation or reduction of $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ on the surviving quartic branch.*

Proof. This is immediate from Theorem 1. □

5. Immediate Structure of the Next Computation

The exact bridge-closure test can now be decomposed into a finite list of concrete burdens.

Definition 5 (Bridge-closure work package)**.** *The* bridge-closure work package *for the surviving quartic branch consists of:*

1. *deriving the full reduced observer equation on the surviving branch after auxiliary elimination*

2. *identifying the exact tensor $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ in* (5)

3. *testing whether the recovered tensor is identically zero, pure gauge/constraint exact, or genuinely observer-accessible*

4. *deciding whether the stable quartic branch should therefore enter the active derivational line or remain only a stable candidate side program.*

Remark 4. *Definition 5 is deliberately narrower than "derive GR from the substrate." The current repository is not yet at that global claim. It is now at the point where the exact remaining tensor to compute has been isolated cleanly.*

6. What This Step Establishes

The present manuscript establishes the following.

1. It records that the same fixed-completion weighted/homogeneous unit-lapse route now has a positive same-class asymptotic/nonlinear-stability closure result.
2. It defines the bridge correction tensor $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ as the exact remaining derivational object on the surviving quartic branch.
3. It distinguishes recovery compatibility and same-class nonlinear stability from true derivational bridge closure.
4. It proves that the next derivational proof-or-failure test is exactly whether $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ vanishes or is only pure gauge/constraint exact.
5. It identifies the explicit bridge-closure work package that should now follow.

It does not establish the following.

1. It does not yet compute the full nonlinear tensor $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$.
2. It does not yet prove that the surviving quartic branch derives the Einstein equation from substrate variables.
3. It does not yet decide whether the surviving quartic branch belongs permanently in the active derivational line.
4. It does not weaken the current strict ordered-flow positivity assumptions or broaden the recovery regime beyond the recorded disciplined scope.

7. Conclusion

The higher-derivative quartic branch has now reached a different kind of next step. The fixed-completion unit-lapse weighted/homogeneous route no longer stops at coercive control, at decay/integrability, or at asymptotic closure. Those steps now exist and close positively. The branch is therefore no longer blocked first by a stability-level obstruction.

That does not mean the derivational program is complete. It means the remaining burden has finally become exact. The question is now whether the reduced observer equation carries any residual branch correction once the surviving branch is written in its proper reduced form. This note records that exact burden by defining the bridge correction tensor $\mathcal{C}^{\mathrm{br}}_{\mu\nu}$ and by stating the proof-or-failure criterion it must satisfy.

At current disciplined scope, that is what we need to do next. If the bridge correction tensor vanishes, or is only gauge/constraint exact, the surviving quartic branch closes onto the exact Einsteinian line in the stated regime. If it does not, then the branch remains a stable recovery-compatible candidate but not a completed substrate derivation. The next manuscript should therefore compute or reduce that tensor directly rather than reopening a stability problem that the present sequence has already cleared.